

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027**

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	2026-27 (ODD)	Course Name	Lab: Hybrid Computing Systems
Semester	7	Course Code	23IOT1704
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	03
Course Outcome	1. Understand Cloud Computing architecture, service models, deployment models, and virtualization. 2. Apply cloud platforms, storage services, and containerization for application deployment. 3. Implement Edge and Fog Computing solutions for IoT applications. 4. Analyze and develop integrated Cloud-Fog-Edge computing applications based on performance evaluation.
Aim	Deployment of a Web Application on a Cloud Platform.
Theory (100 words)	Cloud deployment is the process of hosting and running a web application on a cloud platform so that it can be accessed through the internet. Instead of running the application only on a local computer, cloud services provide remote servers and computing resources for hosting the application. A web application may contain source code, web pages, databases, images, configuration files, and other supporting resources. During deployment, these components are uploaded to or connected with a cloud service. The cloud platform provides resources such as CPU, memory, storage, and networking required to execute the application. The general cloud deployment process involves creating an account on a cloud platform, creating a virtual server or selecting a suitable cloud service, uploading the application files, installing the required software and dependencies, configuring the application, and starting the application. After deployment, the application can usually be accessed using a public IP address, domain name, or URL.

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	Cloud platforms also provide important features such as scalability, availability, security, backup, monitoring, and resource management. Resources can be increased or decreased according to the application requirements. Common cloud platforms include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). Cloud deployment is widely used for websites, APIs, mobile application backends, online services, and business applications.
Procedure and Execution (100 Words)	1. Open the Google Cloud Console and log in with a Google account. 2. Create a new Google Cloud Project for hosting the web application. 3. Complete the billing and account setup if it is required. 4. Navigate to Compute Engine and select the option to create a new Virtual Machine (VM) instance. 5. Choose an appropriate region and zone, and select Ubuntu Linux as the operating system. 6. Select a suitable machine type and configuration according to the requirements of the practical. 7. Create the VM instance and wait until its status changes to Running. 8. Select the SSH option to open the terminal of the cloud VM. 9. Update the installed packages using the appropriate Linux package manager. 10. Install the software required for the web application, such as Python, Node.js, or Apache. 11. Create or upload the required web application files to the VM. 12. Install the necessary libraries, packages, and dependencies required by the application. 13. Start and run the web application on the cloud server. 14. Configure the firewall rules to permit HTTP traffic through the required application port. 15. Copy the External IP address assigned to the VM. 16. Enter the external IP address in a web browser to access the deployed application. 17. Check whether the web application loads correctly in the browser. 18. Test the application by using its main functions and features.



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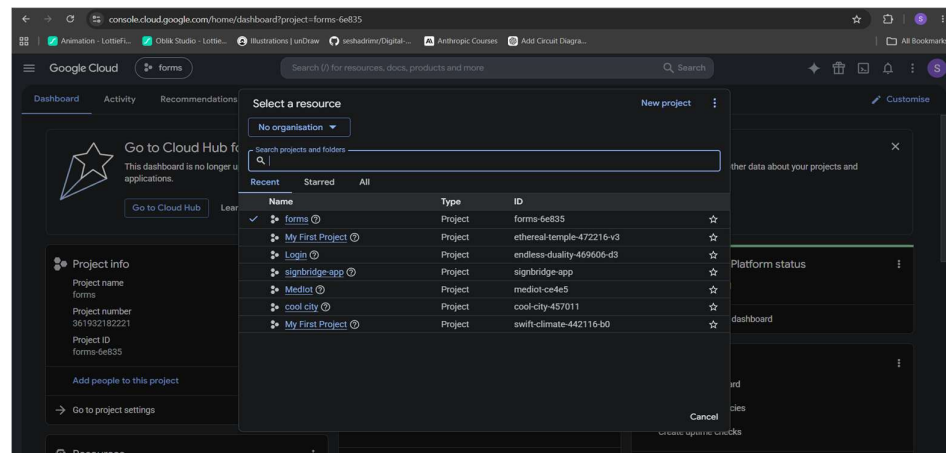
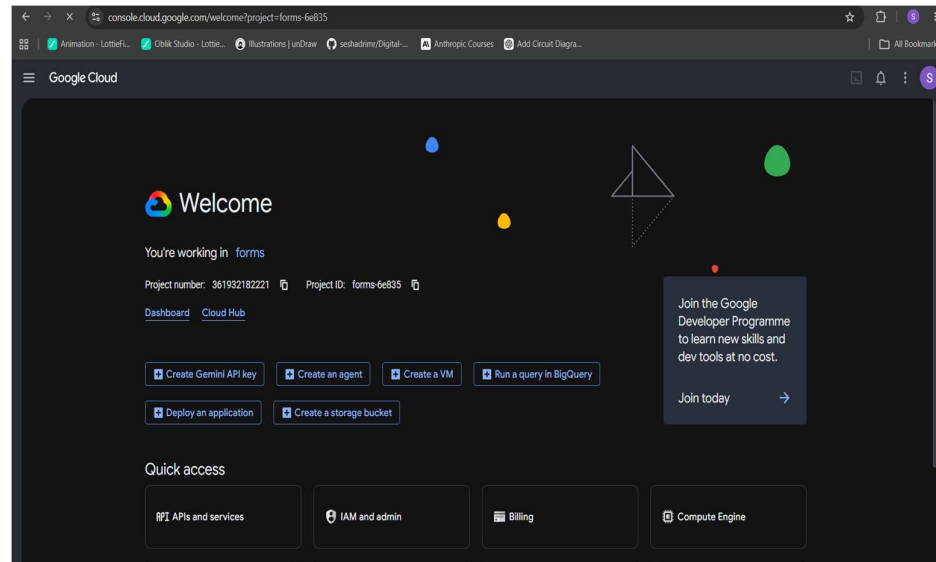
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19. Confirm that the application is accessible through the browser, indicating that the web application has been successfully deployed on Google Cloud Platform.

Stepwise Screenshots with steps:





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New Project

Project name *
Web app

Project ID: web-app-509212. It cannot be changed later. [Edit](#)

Organisation *
sheikhwasimuddin786-org

Select an organisation to attach it to a project. This selection can't be changed later.

Parent resource *
sheikhwasimuddin786-org [Browse](#)

Select the organisation or folder where this resource will be created

[Create](#) [Cancel](#)

Now viewing organisation sheikhwasimuddin786-org



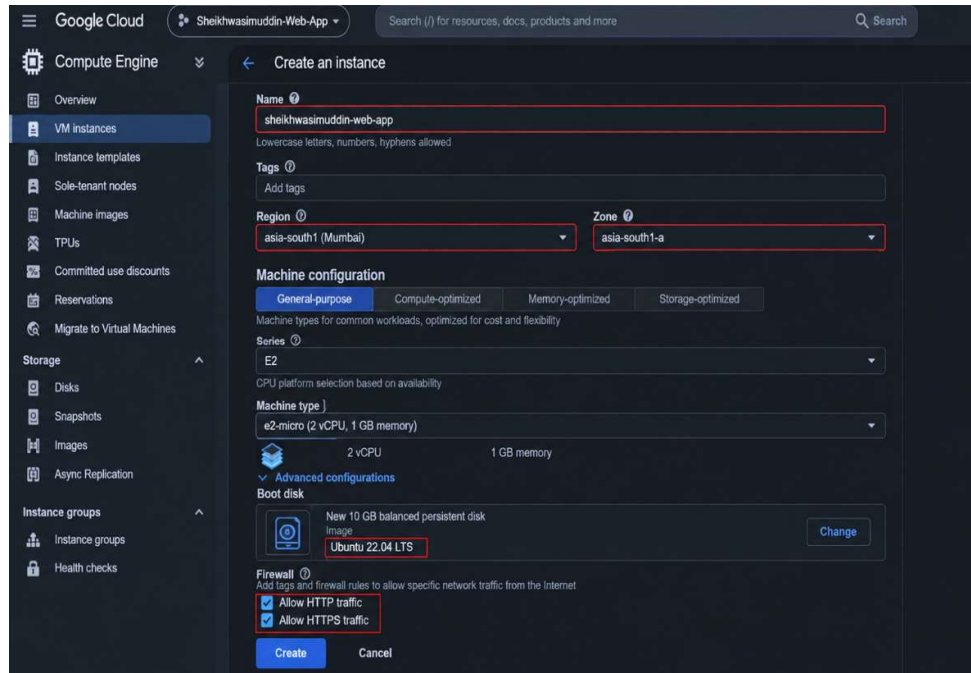
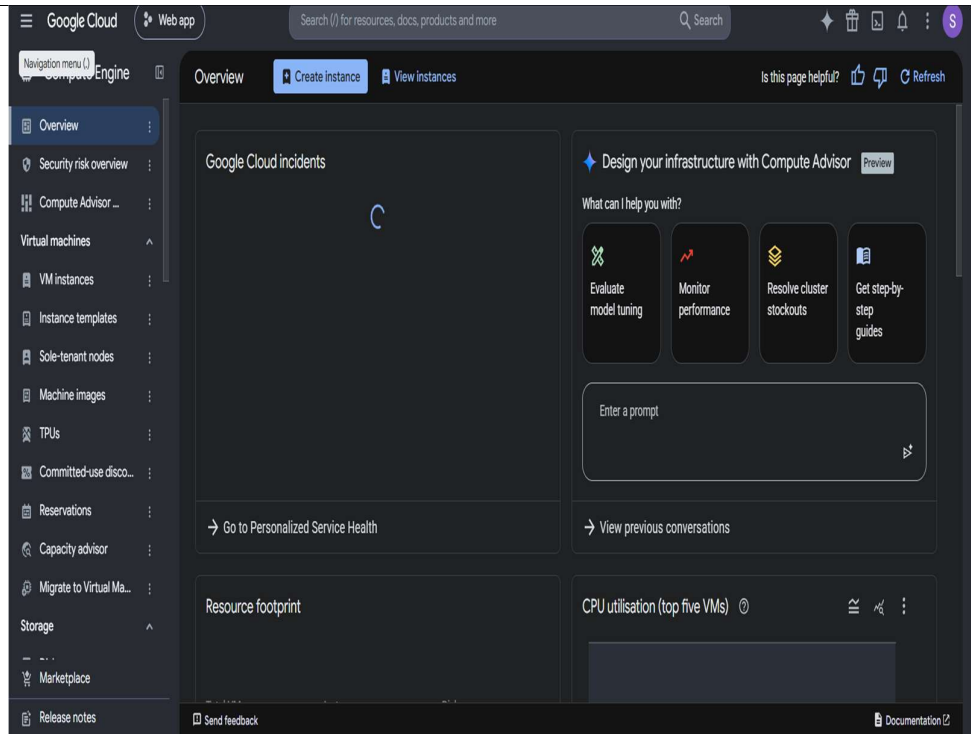
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Conclusion	The web application was successfully deployed on the Google Cloud Platform using a virtual machine. The application was configured on the cloud server and accessed through the internet. This practical helped in understanding the basic process of cloud deployment, server configuration, and web application hosting.
Date	17/07/2026